

# Famous P.D.F. contd.

Uniform P.D.F.  $\rightarrow$

$$E[x^2] = \int_a^b x^2 \cdot f(x) dx$$

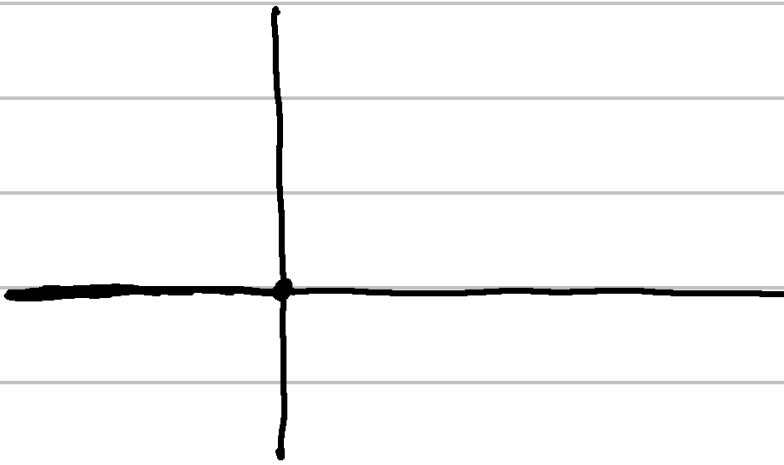
$$= \int_a^b \frac{1}{b-a} \cdot x dx$$

$$= \frac{1}{b-a} \cdot \frac{x^3}{3}$$

$$VAR[x] = \frac{(b-a)^2}{12}$$

# Exponential R.V.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$



$$E[x] = \int_{-\infty}^{\infty} \lambda x \cdot e^{-\lambda x} dx$$

$$\hookrightarrow \lambda \quad [\text{verify}]$$

$$\text{Var}[x] = \frac{1}{\lambda^2}$$

# Gaussian R.V. [Normal R.V.]

$$x \sim f(m, \sigma^2)$$

$$\Omega_x = -\infty, +\infty$$

$$\Omega_x = \mathbb{R}.$$

$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

$$E[x] = m$$

$$\text{Var}[x] = \sigma^2$$

## Central limit theorem

→ The sum of independent R.V.s tends to gaussian distribution.

Remark \* Normality is preserved with linear transformation.

$$X \sim N(m_x, \sigma_x^2)$$

$$Y = aX + b$$

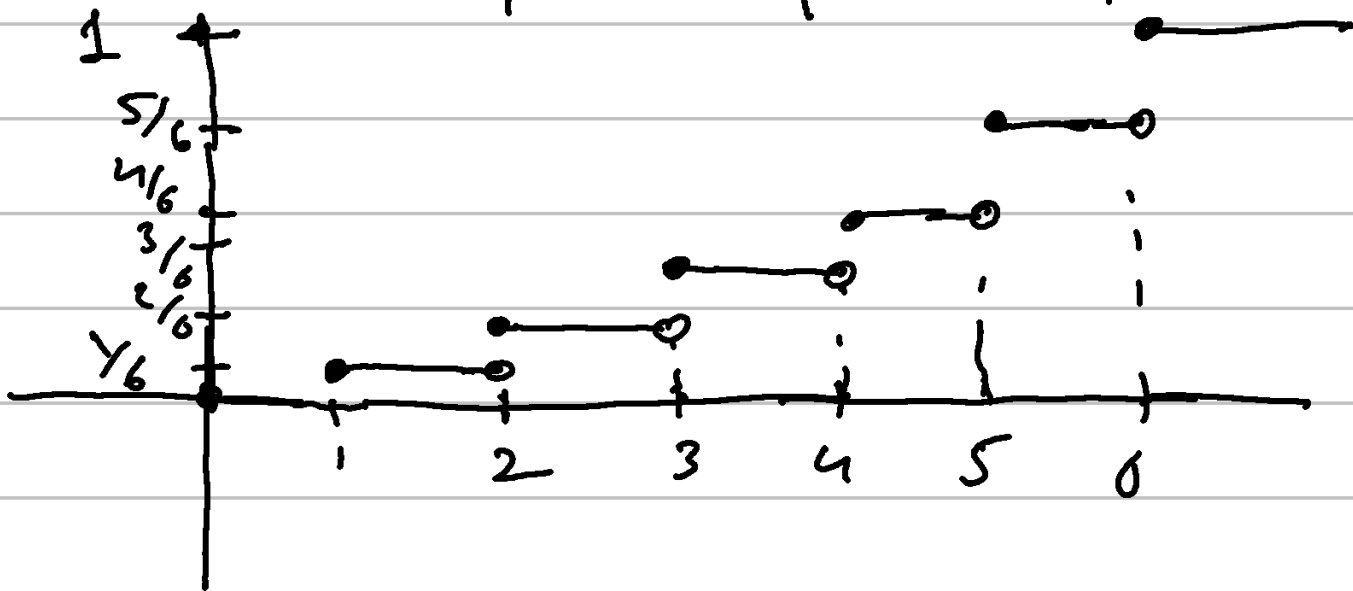
$$Y \sim N(m_y, \sigma_y^2)$$

$$\begin{aligned} m_y &= am_x + b \\ \sigma_y &= a^2 \sigma_x^2 \end{aligned}$$

# Cumulative distribution function CDF

$$F_x(x) = P[X \leq x]$$

ex Plot the cdf a "fair roll of a die"



⇒ CDF is the "primitive" of the PDF.

# Properties of CDF

①  $F_x$  is monotonically non decreasing  
if  $x \leq y \Rightarrow F_x(x) \leq F_x(y)$

②  $F_x(x) \rightarrow 0$  as  $x \rightarrow -\infty$   
 $F_x(x) \rightarrow 1$  as  $x \rightarrow \infty$

③ • If  $X$  is discrete then

$F_x(x)$  is a piecewise const function.

• If  $X$  is cont. then.

$F_x(x)$  is a continuous function of  $x$ .

④ • If  $X$  is discrete

$$F_X(k) = \sum_{i=-\infty}^k P_X(i)$$

$$\begin{aligned} P_X(k) &= P[X \leq k] - P[X \leq k-1] \\ &= F_X[X=k] - F_X[X=k-1] \end{aligned}$$

• If  $X$  is continuous:

$$F_X(x) = \int_{-\infty}^x f_X(u) du$$

$$f_X(x) = \frac{\partial F_X(x)}{\partial x}$$

# Multiple Continuous R.V.

We say that 2 C.R.V. associated with the same experiment are jointly continuous described as

$$P[(x, y) \in B] = \int \int_{(x, y) \in B} f_{x, y}(x, y) dx dy$$

for every subset  $B$  of the 2-d plane

⇒ Marginals

$$\cdot f_x(x) = \int_{-\infty}^{\infty} f_{x, y}(x, y) dy$$

$$\cdot f_y(y) = \int_{-\infty}^{\infty} f_{x, y}(x, y) dx$$



# Derived Distribution

$$Y = g(X) \quad \begin{matrix} \downarrow \\ \text{continuous} \end{matrix}$$

$X \rightarrow \text{C.R.V with PDF } f_X(x)$

$\Rightarrow$  PDF of  $Y$

## ① Principal Method

i) Calculate the CDF of  $Y$ .

$$F_Y(y) = P[Y \leq y] = P[g(X) \leq y]$$

$$\Rightarrow \int_{\{x \mid g(x) \leq y\}} f_X(u) du$$

$$ii) f_Y(y) = F'(y)$$

### Exercise

$$X \rightarrow U[0, 1]$$

$$Y \rightarrow \sqrt{X}$$

$$Q_Y \rightarrow [0, 1]$$

$$\Rightarrow f_X(x) = \begin{cases} 0 & x \geq 1 \\ 1 & 0 < x < 1 \\ 0 & x \leq 0 \end{cases}$$

$$\Rightarrow P[\sqrt{x} \leq y]$$

$$\Rightarrow P[x \leq y^2]$$

$$\int_0^{y^2} f_x(x) dx$$

$$\Rightarrow y^2$$

$$\Rightarrow F_Y(y) = y^2 \quad y \in [0, 1]$$

$$f_Y(y) = 2y$$

→ ② PDF formula for a  $g(-)$  strictly monotonic function and inverse exists.

$$y = g(x) \quad \text{iff} \quad x = h(y)$$

Assuming  $h$  is differentiable

$$f_Y(y) = f_X(h(y)) \cdot \left| \frac{\partial h}{\partial y}(y) \right|$$

Proof using the first method  
and separately for monotonically  
increasing & decreasing.

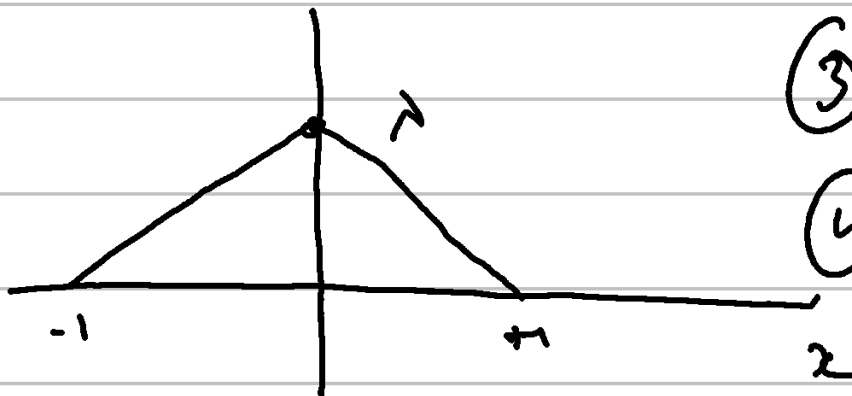
## Exercise

$X \rightarrow$  C.R.V.

① Find  $\lambda$

$$\Omega_x = [-1, +1]$$

②  $f_x(x) = ?$



③  $F_x(x) = ?$

④  $E[X]$

$$\begin{aligned} \textcircled{1} \int_{-1}^1 f_x(x) dx &= \frac{1}{2} \cdot 1 \cdot \lambda + \frac{1}{2} \cdot 1 \cdot \lambda \\ &= \frac{\lambda}{2} + \frac{\lambda}{2} = 1 \\ \boxed{\lambda = 1} \end{aligned}$$

$$(2) \quad f_x(x) = \begin{cases} x+1 & -1 \leq x \leq 0 \\ -x+1 & 0 < x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$(3) \quad F_x(x) = \begin{cases} 0 & x < -1 \\ \frac{x^2-1}{2} + x+1 & -1 \leq x \leq 0 \\ \frac{1}{2} + x - \frac{x^2}{2} & 0 < x \leq 1 \\ 1 & x > 1 \end{cases}$$

$$(4) \quad E[x] = \int_{-1}^1 x f_x(x) dx$$

$$\Rightarrow \int_{-1}^0 x(x+1) dx + \int_0^1 x(1-x) dx$$

$$\Rightarrow \int_{-1}^0 x(x+1) dx + \int_0^1 x(1-x) dx$$

$$\Rightarrow \int_{-1}^0 (x^2 + x) dx + \int_0^1 (x - x^2) dx$$

$$\Rightarrow \left[ \frac{x^3}{3} \right]_{-1}^0 + \left[ \frac{x^2}{2} \right]_{-1}^0 + \left[ \frac{x^2}{2} \right]_0^1 - \left[ \frac{x^3}{3} \right]_0^1$$

$$\Rightarrow \frac{1}{3} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3}$$

$$= 0$$

$$\underline{E[x] = 0}$$

$$\textcircled{4} \quad \text{VAR}[x] = E[x^2] - (E[x])^2$$

$$E[x^2] = \int_{-1}^0 x^2 \cdot (x+1) dx + \int_0^1 x^2 (1-x) dx$$

$$\Rightarrow \frac{1}{6}$$

$$E[x^2] = \frac{1}{6}$$

$$\textcircled{8} \quad P \left[ x < \frac{1}{4}, \quad x^2 \leq \frac{1}{4} \right]$$

$$\frac{1}{2} + x - \frac{x^2}{2}$$

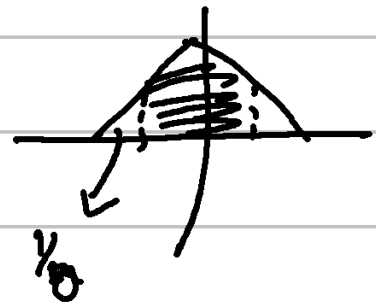
$$\Rightarrow \frac{1}{2} + \frac{1}{4} - \frac{(\frac{1}{4})^2}{2}$$

$$\Rightarrow \frac{1}{2} + \frac{1}{4} - \frac{1}{32} = P[x < \frac{1}{4}]$$

$$P[x^2 \leq \frac{1}{4}]$$

$$P[x < \frac{1}{2} \cap x > -\frac{1}{2}]$$

$$\Rightarrow 1 - \frac{1}{8} \cdot 2 = \underline{\underline{\frac{3}{4}}}$$





$$\frac{3}{8} + \left( P[X < \frac{1}{4}] - \frac{1}{2} \right)$$

$$\Rightarrow \frac{3}{8} - \frac{(\frac{1}{4})^2}{2} + \frac{1}{4} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{2}}$$

$$\Rightarrow \frac{3}{8} - \frac{1}{32} + \frac{1}{4}$$

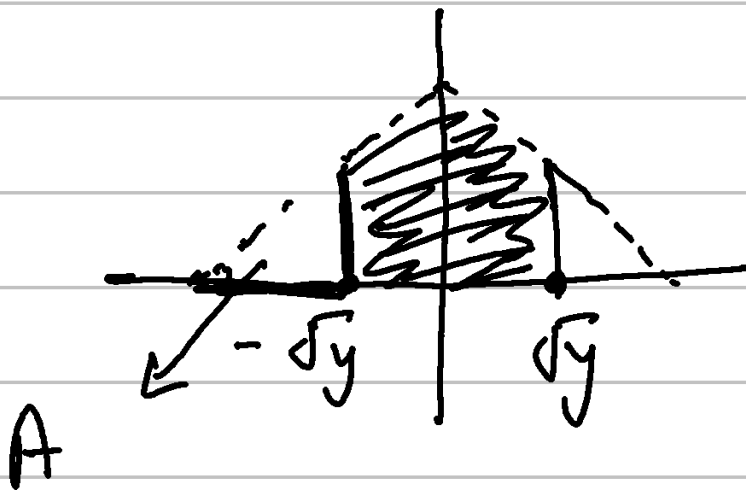
$$\Rightarrow \frac{12 - 1 + 8}{32} = \frac{19}{32}$$

$$\textcircled{6} \quad Y = X^2$$

$$P[Y \leq y] = P[X^2 \leq y]$$

$$\Rightarrow P[|X| \leq \sqrt{y}]$$

$$\Rightarrow P[|x| \leq \sqrt{y}]$$



$$x = -\sqrt{y}$$

$$f_x(x) = 1 - \sqrt{y}$$

$$\begin{aligned} \text{area of } A &= \frac{1}{2} \cdot (1 - \sqrt{y})^2 \\ &= \frac{(1 - \sqrt{y})^2}{2} \end{aligned}$$

$$\text{Excluded area} = 2 \cdot A = (1 - \sqrt{y})^2$$

$$\text{Total area} = 1$$

$$P[|x| \leq \sqrt{y}] = 1 - (1 - \sqrt{y})^2$$

$$\forall y \in [0, 1]$$

$$* F_Y(y) = 1 - (1 - \sqrt{y})^2$$

$$\begin{aligned} f_Y(y) &= +1 \cdot 2(1 - \sqrt{y}) \cdot +\frac{1}{2\sqrt{y}} \\ &= \frac{2}{2} \cdot \frac{1 - \sqrt{y}}{\sqrt{y}} = \frac{1}{\sqrt{y}} - 1 \end{aligned}$$

$$* f_Y(y) = \frac{1}{\sqrt{y}} - 1$$

$$\begin{aligned} E[Y] &= \int_0^1 y \cdot \left( \frac{1}{\sqrt{y}} - 1 \right) dy \\ &= \int_0^1 (\sqrt{y} - y) dy = \left[ \frac{2}{3} y^{3/2} - \frac{y^2}{2} \right]_0^1 \\ &= \frac{2}{3} - \frac{1}{2} = \frac{1}{6} // \end{aligned}$$

## Exercise 2

$$f_X(x) = \begin{cases} K e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

① Under what condition on  $\lambda$  is  $f_X(x)$  a P.D.F

$$\Rightarrow K \int_0^{\infty} e^{-\lambda x} dx = 1$$

$$\boxed{\lambda > 0}$$

$$\Rightarrow K \left[ e^{-\lambda x} \cdot \frac{-1}{\lambda} \right]_0^{\infty} = 1$$

$$\Rightarrow -\frac{K}{\lambda} \cdot (e^{-\infty} - e^0) = 1$$

$$\Rightarrow -\frac{K}{\lambda} \cdot (-1) = 1$$

②  $\boxed{K = \lambda}$

$$\textcircled{3} \quad x, x_0 > 0$$

$$P[X \geq x \mid X > x_0] = \frac{P[X \geq x \cap X > x_0]}{P[X > x_0]}$$

$$\Rightarrow \frac{\int_0^x f_X(u) du}{\int_0^{x_0} f_X(u) du}$$

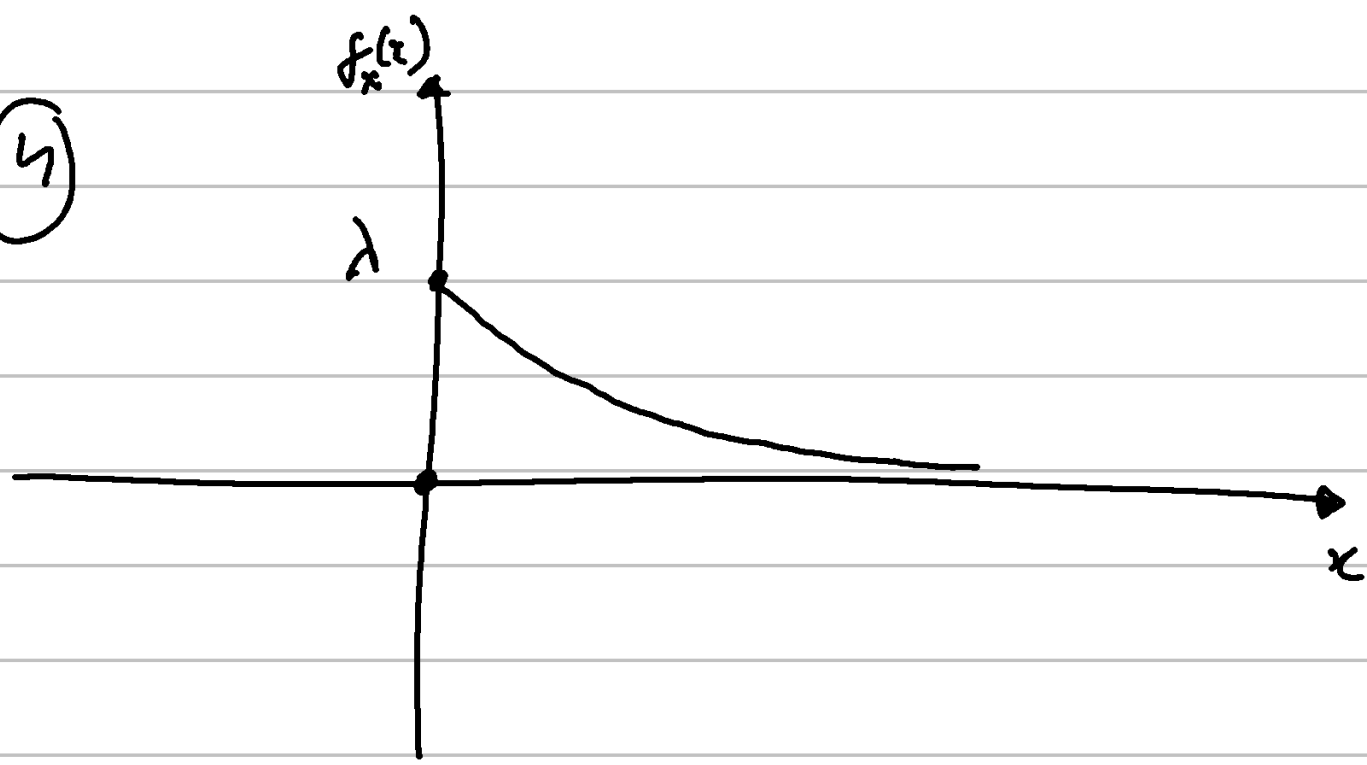
$$\Rightarrow \int_0^x \lambda e^{-\lambda u} du$$

$$\Rightarrow \frac{\left[ -\frac{1}{\lambda} e^{-\lambda u} \right]_0^x}{\left[ -\frac{1}{\lambda} e^{-\lambda u} \right]_0^{x_0}}$$

$$\Rightarrow \frac{e^{-\lambda x}}{e^{-\lambda x_0}}$$

$$\Rightarrow e^{-\lambda(x-x_0)}$$

(4)



⇒ If we consider

$x \rightarrow$  time

$f_x(x) \rightarrow$  Average lifetime after  $x$  time

→ Average lifespan at  $t=0$

$$\boxed{\lambda}$$

→ The probability at any point to survive another  $x_0$  years only depends on  $x_0$

## Exercise 3

$X \rightarrow \text{C.R.V.}$

$$\Omega_X = \mathbb{R}$$

And  $Y \rightarrow \text{R.V.}$

$$X = \ln Y$$

①  $\Omega_Y$ ?

②  $F_Y(y)$ ? (wrt  $F_X(x)$ )

③  $f_Y(y)$ ? (wrt  $f_X(\cdot)$ )

We can assume that  $X$  is gaussian  
 $X \sim N(\mu, \sigma^2)$

④  $E[Y]$ ?

Lets define the median value  $m$  of  $X$  as

$$P[X \geq m] \geq \frac{1}{2} \quad \& \quad P[X \leq m] \geq \frac{1}{2}$$

⑥ Deduce  $m_x$  &  $m_y$

$\Rightarrow$